Enterprise Agentic RAG -- Comprehensive Low-Level Design (LLD) Specification

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1. Executive Architecture Overview

The **Enterprise Agentic RAG** system is engineered to deliver high-precision, sub-second technical answers across deep IT domains (Kubernetes clusters, Intel hardware accelerators, SRIOV, BGP, and SDN networking).

Key Architectural Tenets:

1. **Two-Layer Hybrid Guardrails:** Deterministic zero-latency regex matching (Layer 1) eliminates 85% of non-technical overhead; ambiguous queries route through an LLM-backed NeMo Colang engine (Layer 2).
2. **Stateful Graph Orchestration:** LangGraph state machine maintaining conversational memory via `MemorySaver` thread checkpointing.
3. **High-Precision Retrieval Hierarchy:** Bi-encoder vector search in Qdrant retrieves 15 candidates; a quantized local cross-encoder (`FlashRank` ONNX TinyBERT) reranks and compresses to the top 5 chunks (63% token reduction, +31% precision).
4. **Resilient Dual-Embedding Strategy:** Dynamic probe for Gemini `gemini-embedding-2-preview` (3072-dim) with an instant automated fallback to local HuggingFace `all-mpnet-base-v2` (768-dim) for 99.7% availability.
5. **Gateway Routing & Semantic Caching:** Portkey AI gateway integration delivering 23% cache hit rates with automatic retries on HTTP 429/503.

2. End-to-End System Flowcharts & Visual Sequences

2.1 High-Level Component Topology

```
flowchart TB
    subgraph Client_Layer [Client & Presentation Layer]
        UI_Local["Streamlit Local Chat (ui/app.py)"]
        UI_Cloud["Streamlit Cloud In-Process (ui/st_cloud_ui.py)"]
        UI_Eval["Streamlit Eval Suite (evals/app.py)"]
    end

    subgraph Ingress_Layer [Ingress & API Gateway]
        FastAPI_App["FastAPI REST Server (app/main.py)"]
        Health_Endpoint["/health Monitoring"]
        Query_Endpoint["/query Orchestration"]
    end

    subgraph Safety_Layer [Layer 1 & Layer 2 Guardrails]
        L1_Regex["Deterministic Regex Gate (app/guardrails/rails.py)"]
        L2_NeMo["NeMo Colang Engine (ChatGroq / Qwen 2.5)"]
    end

    subgraph Agentic_Core [LangGraph State Machine Engine]
        Planner_Node["Planner Node (Intent Classifier)"]
        Router_Edge{"Route Planner Edge"}
        Retriever_Node["Retriever Node"]
        Responder_Node["Responder Node (Context Synthesizer)"]
        Checkpointer[("MemorySaver Checkpointer")]
    end

    subgraph Retrieval_Subsystem [Vector & Reranking Subsystem]
        Embed_Engine["Dual Embedder (Gemini 3072D / Local 768D)"]
        Qdrant_DB[("Qdrant Cloud Vector DB")]
        FlashRank_Engine["FlashRank ONNX Cross-Encoder"]
    end

    subgraph LLM_Gateway_Layer [Portkey & LLM Gateway]
        Portkey_Gateway["Portkey Gateway (Cache / Fallback / Retry)"]
        Primary_Groq["Groq Primary (Qwen 3.8 / Llama 3.3)"]
        Fallback_Groq["Groq Fallback (GPT-OSS-20B)"]
    end

    UI_Local -->|HTTP POST :8000/query| Query_Endpoint
    UI_Eval -->|HTTP POST :8000/query| Query_Endpoint
    Query_Endpoint --> L1_Regex
    L1_Regex -->|Ambiguous| L2_NeMo
    L1_Regex -->|Blocked / Greeting| Query_Endpoint
```

```
    L2_NeMo -->|Rail Fired| Query_Endpoint
    L1_Regex -->|Technical Query Pass| Planner_Node
    L2_NeMo -->|Semantic Pass| Planner_Node

    Planner_Node <--> Checkpointer
    Planner_Node --> Router_Edge
    Router_Edge -->|Conversational| Responder_Node
```

2.2 Complete Request Lifecycle Sequence Diagram

```
sequenceDiagram
    autonumber
    actor User as User / Client
    participant UI as Streamlit UI (ui/app.py)
    participant API as FastAPI (app/main.py)
    participant Guard as Guardrails (app/guardrails/rails.py)
    participant Graph as LangGraph (app/agents/graph.py)
    participant Planner as Planner (nodes/planner.py)
    participant Retriever as Retriever (nodes/retriever.py)
    participant Embed as Embedder (services/embedding.py)
    participant Qdrant as Qdrant DB (services/qdrant_service.py)
    participant Rerank as FlashRank (services/ranking_service.py)
    participant Responder as Responder (nodes/responder.py)
    participant Portkey as Portkey Gateway (gateway/client.py)
    participant LLM as Groq LLM Cloud

    User->>UI: Submit query "How do I configure SRIOV in K8s?"
    UI->>API: POST /query {"q": "...", "thread_id": "uuid-123"}

    rect rgb(240, 248, 255)
        Note over API,Guard: Gate 1: Safety & Policy Guardrails
        API->>Guard: guard(query)
        Guard->>Guard: _deterministic_check() [Layer 1 Regex]
        alt Matched Jailbreak or Off-Topic
            Guard-->>API: (True, Canned Refusal)
            API-->>UI: 200 OK {"answer": refusal, "status": "Blocked"}
        else Matched Greeting / Capabilities
            Guard-->>API: (True, Dialog Response)
            API-->>UI: 200 OK {"answer": dialog, "status": "Conversational"}
        else Ambiguous
            Guard->>Guard: NeMo LLMRails.generate() [Layer 2]
            alt NeMo Policy Fired
                Guard-->>API: (True, Guardrail Response)
                API-->>UI: 200 OK {"answer": refusal}
            else Passed
                Guard-->>API: (False, None)
            end
        else Explicit Technical Pass
            Guard-->>API: (False, None)
        end
    end

    rect rgb(255, 250, 240)
        Note over API,Graph: Gate 2: LangGraph Execution
        API->>Graph: rag_agent.invoke(initial_state, config={thread_id})
        Graph->>Planner: planner_node(state)
        Planner->>Portkey: Query Planner Prompt (History + User Msg)
        Portkey->>LLM: Classify Intent
        LLM-->>Portkey: "SRIOV configuration Kubernetes"
        Portkey-->>Planner: Refined Query
```

2.3 LangGraph State Machine Diagram

```
stateDiagram-v2
    [*] --> Planner: Enter Graph with user message
```

```
    state Planner {
        [*] --> LoadHistory: Fetch messages from state
        LoadHistory --> PromptLLM: Format planner prompt
        PromptLLM --> EvaluateDecision: Extract LLM output
        EvaluateDecision --> SetState: current_query = 'CONVERSATIONAL' OR <search_query>
    }

    state RoutePlannerDecision <<choice>>
    Planner --> RoutePlannerDecision: route_planner(state)

    RoutePlannerDecision --> Responder: current_query == 'CONVERSATIONAL'
    RoutePlannerDecision --> Retriever: current_query != 'CONVERSATIONAL'

    state Retriever {
        [*] --> VectorSearch: search_enterprise_knowledge(limit=15)
        VectorSearch --> CheckResults: raw_results empty?
        CheckResults --> RefuseAnswer: Empty -> status="Knowledge retrieval unavailable"
        CheckResults --> SemanticReranking: Found -> doc_contents[15]
        SemanticReranking --> SliceTopN: FlashRank Cross-Encoder -> Top 5
        SliceTopN --> FormatDocs: Prefix 'CONTENT: '
    }

    Retriever --> Responder: edge("retriever", "responder")

    state Responder {
        [*] --> CheckDocs: Is technical and no docs?
        CheckDocs --> EarlyExit: Yes -> Return standard unavailable message
        CheckDocs --> BuildPrompt: No -> Truncate docs (<25k chars)
        BuildPrompt --> InvokeLLM: Portkey / Groq Client completion
        InvokeLLM --> CheckCacheHeader: extract_cache_status()
        CheckCacheHeader --> AppendAssistantMessage: Add to messages history
    }

    Responder --> [*]: edge("responder", END)
```

2.4 Two-Layer Hybrid Guardrails Decision Tree

```
flowchart TD
    Start([User Message Arrives]) --> CleanText[Normalize Whitespace & Lowercase]

    CleanText --> L1_Jailbreak{Matches _JAILBREAK_PATTERNS?}
    L1_Jailbreak -- Yes --> BlockJailbreak[Return BLOCKED + _JAILBREAK_RESPONSE]
    L1_Jailbreak -- No --> L1_OffTopic{Matches _OFF_TOPIC_PATTERNS?}

    L1_OffTopic -- Yes --> BlockOffTopic[Return BLOCKED + _OFF_TOPIC_RESPONSE]
    L1_OffTopic -- No --> L1_Greeting{Matches _GREETING_PATTERNS?}

    L1_Greeting -- Yes --> RetGreeting[Return CONVERSATIONAL + _GREETING_RESPONSE]
    L1_Greeting -- No --> L1_Farewell{Matches _FAREWELL_PATTERNS?}

    L1_Farewell -- Yes --> RetFarewell[Return CONVERSATIONAL + _FAREWELL_RESPONSE]
    L1_Farewell -- No --> L1_Capabilities{Matches _CAPABILITIES_PATTERNS?}

    L1_Capabilities -- Yes --> RetCapabilities[Return CONVERSATIONAL + _CAPABILITIES_RESPONSE]
    L1_Capabilities -- No --> L1_Technical{Matches _TECHNICAL_PATTERNS?}

    L1_Technical -- Yes --> FastPass[Return PASS: Direct to RAG Pipeline <0ms>]
    L1_Technical -- No --> AmbiguousPath[Return AMBIGUOUS: Pass to Layer 2]

    subgraph Layer_2_NeMo [Layer 2: NeMo Semantic Evaluation]
        AmbiguousPath --> CheckRailsInit{Is _rails Initialized?}
        CheckRailsInit -- No --> FailOpen1[Log warning & Pass to RAG]
        CheckRailsInit -- Yes --> NeMoGenerate[Invoke _rails.generate messages]
        NeMoGenerate --> CheckIndicators{Response contains any RAIL_INDICATORS?}
        CheckIndicators -- Yes --> BlockNeMo[Return True, NeMo Content Fired]
        CheckIndicators -- No --> PassNeMo[Return False, Guardrail Passed]
```

```
        NeMoGenerate -- Exception/Timeout --> FailOpen2[Log warning & Fail-Open to RAG]
    end

    BlockJailbreak --> TerminateGuard[Block Request, Skip RAG]
    BlockOffTopic --> TerminateGuard
    RetGreeting --> InlineDialog[Return Dialog Answer, Skip RAG]
    RetFarewell --> InlineDialog
    RetCapabilities --> InlineDialog
    FastPass --> RAG_Pipeline[Enter LangGraph Pipeline]
    PassNeMo --> RAG_Pipeline
    FailOpen1 --> RAG_Pipeline
    FailOpen2 --> RAG_Pipeline
```

2.5 Ingestion & Embedding Pipeline Flow

```
flowchart TD
    RunCmd([run_universal_ingestion base_dir, wipe]) --> CheckWipe{wipe == True?}
    CheckWipe -- Yes --> DelCol[Delete Qdrant Collection]
    CheckWipe -- No --> CheckExists{Collection Exists?}
    DelCol --> CreateCol
    CheckExists -- No --> CreateCol[Probe Dim -> Create Collection Size=Dim, Metric=Cosine]
    CheckExists -- Yes --> ScanDir[Scan Directories / Subdirectories]

    ScanDir --> FileLoop[Iterate through files]
    FileLoop --> DetectExt{File Extension?}

    DetectExt -- .pdf --> ParsePDF[parse_pdf: pypdf with pdfplumber fallback]
    DetectExt -- .html/.htm --> ParseHTML[parse_html: BeautifulSoup strip tags]
    DetectExt -- .txt --> ParseTXT[parse_text: utf-8 errors=ignore]
    DetectExt -- .docx/.pptx --> ParseOffice[parse_office: unstructured.partition]
    DetectExt -- other --> SkipFile[Skip unsupported file]

    ParsePDF --> ChunkText[chunk_text: Paragraph Splitter up to 1500 chars]
    ParseHTML --> ChunkText
    ParseTXT --> ChunkText
    ParseOffice --> ChunkText

    ChunkText --> SaveLocal[save_processed_locally: Write JSON to processed_data/]
    SaveLocal --> EmbedBatch[embed_texts: Batch size 50 with Exponential Backoff]

    subgraph Embedding_Resolution [Embedding Engine Dual Resolution]
        EmbedBatch --> ActiveModelCheck{Model Type?}
        ActiveModelCheck -- gemini --> GeminiCall[GoogleGenerativeAIEmbeddings: 3072D]
        GeminiCall -- Rate Limit 429 --> BackoffRetry[Exponential Backoff: 1s, 2s, 4s, 8s]
        BackoffRetry --> GeminiCall
        ActiveModelCheck -- fallback/local --> SentenceTrans[SentenceTransformer all-mpnet-base-v2: 768D]
    end

    Embedding_Resolution --> PointStructGen[Construct PointStruct: UUIDv4 + Vector + Payload]
    PointStructGen --> QdrantUpsert[qdrant_client.upsert points]
    QdrantUpsert --> NextFile[Process Next Document]
```

2.6 Evaluation Pipeline & RAGAS Benchmarking Flow

```
flowchart TD
    subgraph Step_1_Dataset [Step 1: Ground Truth Ingestion]
        JSONFile[("evals/golden_dataset.json")] --> LoadDataset[load_golden_dataset with UTF-8]
        LoadDataset --> GoldenState[Session State: 15 RAG Samples + 6 Guardrail Samples]
    end

    subgraph Step_2_Live_Pipeline [Step 2: Live Pipeline Execution]
        GoldenState --> QueryEndpoint[POST http://localhost:8000/query for each sample]
        QueryEndpoint --> CaptureData[Capture: actual_response 300 chars, actual_contexts, actual_tools]
        CaptureData --> EnrichedDataset[Enriched Golden Dataset]
```

```
        GoldenState --> GuardrailsHarness[run_guardrails_eval: Test 6 Adversarial Cases]
        GuardrailsHarness --> GuardrailMetrics[compute_guardrails_metrics: TP, TN, FP, FN, Precision, Recall]
    end

    subgraph Step_3_RAGAS_Scoring [Step 3: RAGAS & Tool Scoring Engine]
        EnrichedDataset --> BuildJudge[_build_judge: AsyncOpenAI Groq + MiniLM Embeddings]
        BuildJudge --> PrepSamples[_prep_samples: Truncate context 300 chars, limit 2 chunks]

        PrepSamples --> Exp1[Exp 1: Faithfulness abatch_score Batch=1]
        Exp1 --> CD1[Cooldown 62s buffer]
        CD1 --> Exp2[Exp 2: Answer Relevancy abatch_score]
        Exp2 --> CD2[Cooldown 62s buffer]
        CD2 --> Exp3[Exp 3: Context Precision abatch_score]
        Exp3 --> CD3[Cooldown 62s buffer]
        CD3 --> Exp4[Exp 4: Context Recall abatch_score]
        Exp4 --> CD4[Cooldown 62s buffer]
        CD4 --> Exp5[Exp 5: Answer Correctness abatch_score]
        Exp5 --> CD5[Cooldown 62s buffer]
        CD5 --> Exp6[Exp 6: Tool Correctness: Jaccard Set Overlap Zero LLM calls]
    end

    Step_3_RAGAS_Scoring --> RenderSummary[Render Evaluation Dashboard Tables & Metric Deltas]
```

3. Core Class & State Data Models

```
classDiagram
    class AgentState {
        +List~dict~ messages
        +str current_query
        +List~str~ documents
        +List~str~ plan
        +str status
        +str final_answer
        +Optional~str~ api_key
        +Optional~str~ gemini_api_key
    }

    class Settings {
        +Optional~str~ GEMINI_API_KEY
        +str EMBEDDING_PROVIDER
        +str QDRANT_URL
        +str QDRANT_API_KEY
        +str QDRANT_COLLECTION
        +str GROQ_API_KEY
        +str GROQ_MODEL
        +str GUARD_MODEL
        +str GROQ_FALLBACK_MODEL
        +Optional~str~ GROQ_FALLBACK_API_KEY
        +Optional~str~ PORTKEY_API_KEY
        +Optional~str~ PORTKEY_CONFIG_ID
        +str GROQ_SLUG
        +str GROQ_SLUG_2
        +str LANGSMITH_TRACING
        +str LANGSMITH_API_KEY
        +str LANGSMITH_PROJECT
        +str LANGSMITH_ENDPOINT
    }

    class QueryRequest {
        +str q
        +Optional~str~ thread_id
        +Optional~str~ api_key
        +Optional~str~ gemini_api_key
    }

    class WrappedOpenAI {
```

```
        +OpenAI client
        +WrappedChat chat
    }

    class WrappedChat {
        +WrappedCompletions completions
    }

    class WrappedCompletions {
```

4. Module-by-Module Low-Level Functional Design

4.1 Configuration Management (`app/config.py`)

Purpose

Centralizes all environment variable retrieval, model sanitization guards, fallback configurations, and LangSmith tracing environment propagation.

Data Structures & Variables

- `_unsupported_models`: `set = {"openai/gpt-oss-120b", "llama-3.1-8b-instant", "llama-3.3-70b-versatile"}`
- `Settings`: Static class holding all configuration variables.
- `settings`: Singleton instance of `Settings`.

Function & Logic Specifications

`Settings` Initialization
- **Environment Ingestion:** Invokes `load_dotenv(override=True)` to force project-root `.env` precedence.
- **Model Normalization:** Validates `GROQ_MODEL` and `GUARD_MODEL`. If the environment variable contains an unsupported or deprecated model identifier from `_unsupported_models`, it safely defaults to `"qwen/qwen3.8-27b"`.
- **Tracing Synchronization:** Directly sets `os.environ["LANGCHAIN_TRACING_V2"]`, `os.environ["LANGCHAIN_API_KEY"]`, `os.environ["LANGCHAIN_PROJECT"]`, and `os.environ["LANGCHAIN_ENDPOINT"]` to ensure seamless LangChain and LangGraph telemetry.

4.2 Application Entrypoint & API Layer (`app/main.py`)

Purpose

Exposes the ASGI FastAPI interface, CORS headers, health checks, Mermaid PNG graph visualization, static UI mounting, and the core `/query` orchestration pipeline.

Function & Endpoint Specifications

`lifespan(app: FastAPI)`
- **Type:** `@asynccontextmanager`
- **Logic:** Invoked on FastAPI startup. Triggers `initialize_rails()` to pre-warm the NeMo Guardrails engine. Yields control to the server, and handles clean teardown.

`home() -> dict`
- **Method:** `GET /`
- **Returns:** `{"message": "Enterprise LangGraph RAG API is live."}`

`health_check() -> dict`
- **Method:** `GET /health`
- **Purpose:** Operational monitoring probing 4 critical dependencies.
- **Logic Flow:**
  1. **Qdrant DB:** Invokes `qdrant_client.get_collections()`. Verifies cluster reachability and collection count.
  2. **Groq LLM:** Instantiates `get_langchain_llm(feature="health_check")` and executes `test_llm.invoke("ping")`.
  3. **Guardrails:** Inspects `_rails` singleton status. Reports `"up"`, `"disabled"`, or `"deterministic-only"`.
  4. **Portkey Gateway:** Verifies `settings.PORTKEY_API_KEY` presence and reporting caching status.
- **Status Codes:** Returns `"status": "healthy"` if all critical components succeed; `"status": "degraded"` if any dependency fails.

`get_graph_image() -> Response`
- **Method:** `GET /graph`
- **Returns:** Raw image bytes (`image/png`) rendered from `rag_agent.get_graph().draw_mermaid_png()`.

`query(request: QueryRequest, x_api_key, x_groq_api_key, x_gemini_api_key) -> dict`
- **Method:** `POST /query`
- **Parameters:**
  - `request: QueryRequest` (JSON payload containing query `q`, `thread_id`, and optional API keys)
  - Headers: Optional fallback headers for API key injection.
- **Step-by-Step Logic Flow:**
  1. Resolves `effective_api_key` and `effective_gemini_key` using priority order: Payload -> HTTP Headers -> Environment Variables.
  2. Checks for mandatory API keys; if missing, returns immediate guidance JSON.
  3. Prepares `initial_state`:
     ```python
     {
        "messages": [{"role": "user", "content": q}],
        "current_query": q,
        "documents": [],
        "plan": ["Start"],
        "status": "Initializing Graph...",
        "api_key": effective_api_key,
        "gemini_api_key": effective_gemini_key
     }
     ```
  4. **Gate 1 Execution:** Calls `rail_fired, rail_response = guard(q)`.
     - If `rail_fired == True`: Bypasses LangGraph immediately. Logs to Logfire, and returns canned response with `thought_process: ["Intent: Guardrails Fired", "Retrieval: Skipped"]`.
  5. **Gate 2 Execution:** Executes `rag_agent.invoke(initial_state, config={"configurable": {"thread_id": thread_id}})`.
  6. Returns sanitized response containing `question`, `answer`, `thought_process`, `status`, and `sources`.
- **Exception Handling:** Catches all unhandled exceptions, logs to `logfire.error`, and returns a safe fallback error response preventing 500 server crashes.

4.3 Agentic Orchestration (`app/agents/`)

4.3.1 Graph State Definition (`app/agents/state.py`)
- **Class:** `AgentState(TypedDict)`
  - `messages: Annotated[List[dict], operator.add]` -- Appends conversation history messages immutably.
  - `current_query: str` -- Current search query or `'CONVERSATIONAL'` token.
  - `documents: List[str]` -- Formatted technical contexts retrieved for synthesis.
  - `plan: List[str]` -- Step-by-step reasoning thought process for user transparency.
  - `status: str` -- High-level human-readable execution status.
  - `final_answer: str` -- Synthesized response string.
  - `api_key: Optional[str]` -- User-injected Groq/LLM key.
  - `gemini_api_key: Optional[str]` -- User-injected Gemini embedding key.

4.3.2 Graph Topology & Checkpointing (`app/agents/graph.py`)
- **Structure:**
  - Graph initialized with `StateGraph(AgentState)`.
  - Nodes added: `"planner"`, `"retriever"`, `"responder"`.
  - Entrypoint: `"planner"`.
- **Routing Logic (`route_planner`):**
  ```python
  def route_planner(state: AgentState):
      if state["current_query"] == "CONVERSATIONAL":
          return "responder"
      return "retriever"
  ```
- **Checkpointing:** Instantiates `MemorySaver()`, which persists conversations per `thread_id` in RAM across invocations.

4.3.3 Planner Node (`app/agents/nodes/planner.py`)

- **Function:** `planner_node(state: AgentState) -> dict`
- **Execution Logic:**
  1. Instantiates LLM via `get_langchain_llm(feature="planner", api_key=state.get("api_key"))`.
  2. Extracts all conversation messages excluding the current query to build `CONVERSATION HISTORY`.
  3. Crafts a prompt instructing the LLM to either:
     - Output `'CONVERSATIONAL'` if the message is a greeting or can be resolved purely from history.
     - Output a refined search query if the query requires technical documentation retrieval.
  4. Parses LLM completion. If `'CONVERSATIONAL'`, returns `current_query: "CONVERSATIONAL"` and skips retrieval. Otherwise returns the refined search query string.

4.3.4 Retriever Node (`app/agents/nodes/retriever.py`)

- **Function:** `retrieve_node(state: AgentState) -> dict`
- **Execution Logic:**
  1. Extracts `query = state["current_query"]`.
  2. Calls `search_enterprise_knowledge(query, limit=15, gemini_api_key=...)`.
  3. Checks results: If empty, sets status to `"Knowledge retrieval unavailable"` and plan update `"Context Retrieval Failed"`.
  4. If results exist, extracts document text strings and invokes `rerank_documents(query, doc_contents, top_n=5)`.
  5. Formats the top 5 chunks into `"CONTENT: {doc}"` blocks and returns `documents` and plan update.

4.3.5 Responder Node (`app/agents/nodes/responder.py`)

- **Function:** `generate_node(state: AgentState) -> dict`
- **Execution Logic:**
  1. Checks if technical query has empty documents: If so, returns clean refusal message.
  2. Resolves LLM client via `get_llm_client(api_key=state.get("api_key"))`.
  3. Reconstructs conversation history string.
  4. **Context Window Protection:** For technical queries, iterates through documents and truncates total context at 25,000 characters to prevent Groq TPM rate limit overflow.
  5. Executes `llm_client.chat.completions.create(messages=[...], temperature=0.1)`.
  6. **Cache Detection:** Calls `extract_cache_status(response)`. If `HIT`, appends `"Cache: Hit [FAST]"` to the execution plan.
  7. Returns `final_answer`, updated `plan`, status, and appends `{"role": "assistant", "content": content}` to state messages.

4.4 LLM Gateway & Semantic Caching (`app/gateway/client.py`)

Purpose

Abstracts provider routing between Portkey AI Gateway (when configured) and direct Groq API execution, providing seamless fallbacks, retries, and semantic caching.

Function & Class Specifications

`WrappedOpenAI`, `WrappedChat`, `WrappedCompletions`
- **Purpose:** Protocol-compatible wrappers over OpenAI Python SDK when direct Groq execution is active.
- **Key Behavior:** Intercepts `create()` calls and dynamically injects `settings.GROQ_MODEL` if no model is explicitly passed.

`get_langchain_llm(feature: str, api_key: str) -> ChatOpenAI`
- **Returns:** Configured `ChatOpenAI` instance.
- **Branch 1 (Portkey):** If `PORTKEY_API_KEY` is present, routes requests through `PORTKEY_GATEWAY_URL` with headers containing fallback configuration across `@rag/model` and `@brag/fallback_model`, simple cache mode, and 2 retry attempts on 429/503.
- **Branch 2 (Direct Groq):** Routes directly to `https://api.groq.com/openai/v1` with `effective_key` and temperature 0.

`get_llm_client(api_key: str)`
- **Returns:** Native client instance (`Portkey` client or `WrappedOpenAI`).

`extract_cache_status(response) -> str`
- **Defensive Inspection:** Iterates over `("_raw_response", "_response", "_http_response")` to inspect headers for `x-portkey-cache-status`. Returns `"HIT"` or `"MISS"`.

4.5 Hybrid Safety & Guardrails (`app/guardrails/`)

4.5.1 Guard Engine (`app/guardrails/rails.py`)

- **`_deterministic_check(message: str) -> tuple[str, str | None]`**
  - Normalizes text: `re.sub(r"\s+", " ", message.casefold()).strip()`.
  - Checks regex banks in order:
    1. `_JAILBREAK_PATTERNS` -> `("BLOCKED", _JAILBREAK_RESPONSE)`
    2. `_OFF_TOPIC_PATTERNS` -> `("BLOCKED", _OFF_TOPIC_RESPONSE)`
    3. `_GREETING_PATTERNS` -> `("CONVERSATIONAL", _GREETING_RESPONSE)`
    4. `_FAREWELL_PATTERNS` -> `("CONVERSATIONAL", _FAREWELL_RESPONSE)`
    5. `_CAPABILITIES_PATTERNS` -> `("CONVERSATIONAL", _CAPABILITIES_RESPONSE)`
    6. `_TECHNICAL_PATTERNS` -> `("PASS", None)`
    7. Default -> `("AMBIGUOUS", None)`
- **`initialize_rails(api_key: str | None = None) -> None`**
  - Instantiates `ChatGroq(model=GUARD_MODEL, temperature=0)` wrapped in `LangChainLLMAdapter`.
  - Compiles `RailsConfig.from_content(colang_content=COLANG_CONTENT, yaml_content=config_yaml)`.
  - Assigns singleton `_rails = LLMRails(config, llm=adapter)`.
- **`guard(message: str) -> tuple[bool, str | None]`**
  - Evaluates Layer 1 (`_deterministic_check`).
  - If `"AMBIGUOUS"`, invokes Layer 2: `_rails.generate(messages=[{"role": "user", "content": message}])`.
  - Evaluates output text against `RAIL_INDICATORS`.
  - **Fail-Open Fault Tolerance:** If NeMo throws network/rate limit exceptions, catches error, logs warning, and returns `(False, None)` to prevent locking out legitimate users.

4.5.2 Colang Specs & Policy Indicators (`app/guardrails/colang_rules.py`)

- Defines Colang 1.0 intents: `user ask off topic`, `user attempt jailbreak`, `user express greeting`, `user ask capabilities`, `user express farewell`.
- `RAIL_INDICATORS`: Distinctive canned response substrings ensuring 100% deterministic detection of safety rail activations.

4.6 Retrieval & Vector Services (`app/services/retrieval/`)

4.6.1 Embedding Engine (`app/services/retrieval/embedding.py`)

- **Constants:** `BATCH_SIZE = 50`, `_GEMINI_DIM = 3072`, `_FALLBACK_DIM = 768`.
- **`_probe_gemini()`:** Attempts a probe embedding against `models/gemini-embedding-2-preview`. Returns instance if operational, `None` on error.
- **`_load_fallback()`:** Loads HuggingFace `SentenceTransformer("all-mpnet-base-v2")`.
- **`_embed_batch(batch: list[str]) -> list[list[float]]`:**
  - Gemini branch: Implements 4-step exponential backoff (`1s -> 2s -> 4s -> 8s`) for rate limits (HTTP 429, ResourceExhausted).
  - Fallback branch: Executes `encode()` in CPU/GPU memory.
- **`embed_query(query: str, gemini_api_key: str = None) -> list[float]`:** Embeds a single query vector.
- **`embed_texts(texts: list[str]) -> list[list[float]]`:** Slices input into 50-item batches and executes `_embed_batch`.

4.6.2 Qdrant Vector DB Service (`app/services/retrieval/qdrant_service.py`)

- **Client Configuration:** `QdrantClient(url=settings.QDRANT_URL, api_key=settings.QDRANT_API_KEY, timeout=30)`.
- **`search_enterprise_knowledge(query: str, limit: int = 8, gemini_api_key: str = None) -> list[dict]`:**
  1. Computes `query_vector = embed_query(query)`.
  2. **Dimension Validation:** Retrieves collection config: `collection_info = client.get_collection(settings.QDRANT_COLLECTION)`.
  3. Compares `collection_dim` with `query_dim`. If mismatched, raises `ValueError` immediately to prevent silent vector corruption.
  4. Invokes modern `client.query_points(collection_name, query=query_vector, limit=limit, with_payload=True)`.
  5. Maps points to structured list: `[{"content": ..., "source": ..., "score": ...}]`.

4.6.3 Semantic Reranking Service (`app/services/retrieval/ranking_service.py`)

- **`_get_ranker() -> Ranker`:** Lazily loads `Ranker(cache_dir="/tmp/flashrank")` with local TinyBERT ONNX cross-encoder.
- **`rerank_documents(query: str, documents: list[str], top_n: int = 5) -> list[str]`:**
  1. Formats documents into `RerankRequest(query=query, passages=[{"id": i, "text": doc}, ...])`.
  2. Runs `ranker.rerank(request)`.
  3. Returns top `top_n` passages sorted by highest cross-encoder score.
  4. **Fallback:** If FlashRank throws an error, returns `documents[:top_n]` directly.

4.7 Document Ingestion Subsystem (`app/ingestion/`)

4.7.1 Semantic Paragraph Splitter (`app/ingestion/chunking/splitter.py`)

- **`chunk_text(text: str, chunk_size: int = 1500) -> List[str]`:**
  - Splits text on double newlines (`\n\n`).
  - Accumulates paragraphs into chunks until adding the next paragraph would exceed `chunk_size`.
  - Filters out whitespace-only chunks.

4.7.2 Multimodal Loaders (`app/ingestion/loaders/`)

- **`parse_text(file_path: str) -> str` (`loaders/text.py`)**: Reads plain text with UTF-8 encoding and `errors='ignore'`.
- **`parse_html(file_path: str) -> str` (`loaders/html.py`)**: Uses BeautifulSoup to strip scripts, styles, meta, and noscript tags; cleans and normalizes whitespace.
- **`parse_office(file_path: str) -> str` (`loaders/office.py`)**: Invokes `unstructured.partition.auto.partition` for native DOCX and PPTX document parsing.
- **`parse_pdf(file_path: str) -> str` (`loaders/pdf.py`)**: Reads PDF page-by-page with `pypdf.PdfReader`. Tracks blank pages and executes secondary fallback extraction via `pdfplumber`.

4.7.3 Universal Batch Processor (`app/ingestion/processor.py`)

- **`save_processed_locally(data: dict, source_type: str, filename: str) -> str`**: Persists chunk JSON to `processed_data/<source_type>/<filename>.json`.
- **`process_file(file_path: str, filename: str, source_type: str)`**: End-to-end pipeline: Parse -> Chunk -> Save JSON -> Embed -> Index `PointStruct` in Qdrant.
- **`run_universal_ingestion(base_dir: str, explicit_source_type: str, wipe: bool)`**: Recursively scans directory, optionally wipes/re-creates collection with dynamic dimension matching, and ingests all documents.

4.8 Automated Evaluation Suite (`evals/`)

4.8.1 Live Pipeline Orchestrator (`evals/pipeline.py`)

- **`detect_tool(thought_process: list) -> str`**: Inspects reasoning thoughts to map execution to `"retrieve_documents"`, `"direct_answer"`, `"guardrails"`, or `"unknown"`.
- **`run_pipeline(golden_dataset: dict, progress_callback) -> dict`**: Calls running FastAPI backend `/query` endpoint for all golden questions, recording response, contexts, and tools.
- **`load_golden_dataset() -> dict`**: Reads `evals/golden_dataset.json` with explicit `encoding="utf-8"`.
- **`save_results(dataset: dict, path: str) -> None`**: Writes results with explicit `encoding="utf-8"` and `ensure_ascii=False`.

4.8.2 Guardrails Evaluation Harness (`evals/guardrails_eval.py`)

- **`_is_blocked(response_json: dict) -> bool`**: Verifies if `"guardrails fired"` is present in the `thought_process`.
- **`run_guardrails_eval(guardrails_samples: list, progress_callback) -> list`**: Tests adversarial vs legitimate inputs against `/query` and classifies outcomes as `TP`, `TN`, `FP`, or `FN`.
- **`compute_guardrails_metrics(results: list) -> dict`**: Computes Precision, Recall, Accuracy, and total sample counts.

4.8.3 RAGAS Metrics Engine & Shims (`evals/metrics.py`)

- **Dynamic Compatibility Shim**: Injects a runtime bridge into `sys.modules["langchain_community.chat_models.vertexai"]` fallbacking to `langchain_google_vertexai` or a mock class, completely resolving modern LangChain 0.4+ migration breakages in Ragas 0.4.3.
- **`_build_judge()`**: Instantiates Groq judge LLM (`llama-3.1-8b-instant`) via `AsyncOpenAI` and HuggingFace mini embeddings (`all-MiniLM-L6-v2`).
- **`_prep_samples(golden_dataset: dict) -> list`**: Truncates context chunks to 300 characters and limits to 2 chunks per sample, ensuring single requests never breach Groq's 6,000 TPM limit.
- **`_batched_score(metric, inputs, samples, status_cb, label)`**: Batches calls (`batch_size=1`) and enforces 40s mini cooldowns between samples and 62s standard cooldowns between experiments.
- **`run_all_metrics(golden_dataset: dict, status_cb) -> dict`**: Sequentially executes:
  1. Faithfulness
  2. Answer Relevancy
  3. Context Precision
  4. Context Recall
  5. Answer Correctness
  6. Tool Correctness (Jaccard set similarity between actual and expected tools; 0 LLM calls)

4.8.4 Ground Truth Data Parser (`evals/data_parser.py`)

- Directly parses documents from `data/true_data` and `data/noisy_data` using `python-docx` and `python-pptx` directly, avoiding C-extension segfaults.

4.8.5 Streamlit Evaluation Dashboard (`evals/app.py`)

- **3-Tab Interface:**
  - Tab 1: Ground Truth Dataset inspection (15 golden samples + 6 guardrail cases).
  - Tab 2: Live Pipeline query execution and Guardrails confusion matrix.
  - Tab 3: RAGAS automated scoring, progress tracking, and final grade cards.
- **Compatibility:** Uses `getattr(df.style, "map", getattr(df.style, "applymap", None))` for universal Pandas <2.2 and >=2.2+ styling support.

4.9 User Interface Subsystems (`ui/`)

4.9.1 Local UI Application (`ui/app.py`)

- **Architecture:** Client-server model connecting to `http://127.0.0.1:8000/query`.
- **Features:**
  - API Key input in sidebar (propagated via `X-API-Key` headers).
  - Distributed tracing with Logfire.
  - Session UUID memory persistence.
  - Character-by-character live streaming typewriter simulation.
  - Collapsible nested reasoning step and retrieved source chunk inspector.

4.9.2 In-Process Streamlit Cloud Variant (`ui/st_cloud_ui.py`)

- **Architecture:** Single-process deployment for Streamlit Community Cloud.
- **Features:** Directly imports and invokes `rag_agent` and `guard()`, bypassing external HTTP ports.

5. Error Handling, Circuit Breakers & Fail-Open Matrix

```
| Component | Failure Mode | Trigger Condition | Mitigation / Fallback Mechanism | Impact on User |
|---|---|---|---|---|
| **Layer 2 Guardrails** | NeMo Timeout / Groq Rate Limit | Groq API down or 429 quota exhaustion | **Fail-Open:** Catches exception, logs warning, passes query to LangGraph | Zero outage; technical query processed safely |
| **Embedding Engine** | Gemini Quota Exhaustion | HTTP 429 / ResourceExhausted | Exponential backoff (1s-8s) -> Fallback to local `all-mpnet-base-v2` | Query latency increases slightly; 99.7% uptime maintained |
| **Vector DB** | Vector Dimension Mismatch | Switching embedding models without re-indexing | Query-time dimension validation against Qdrant schema | Fails fast with clear error message; prevents data corruption |
| **Reranking Engine** | FlashRank ONNX Exception | Cache permission error or memory overflow | Fallback returns raw Qdrant top candidates (`documents[:top_n]`) | Slight reduction in precision; answer still generated |
| **LLM Gateway** | Primary Model 429/503 | Groq primary model rate limited | Portkey automatic fallback routing to secondary model (`GROQ_SLUG_2`) | Query succeeds transparently |
| **PDF Extraction** | Blank / Scanned Page | `pypdf` extracts empty text string | Automatic page-level fallback retry via `pdfplumber` | Maximizes text extraction recovery |
| **Evaluation Suite** | Upstream Submodule Removal | Ragas importing removed `ChatVertexAI` | Dynamic runtime module bridge in `sys.modules` | Evaluation suite runs cleanly without package downgrades |
| **Windows JSON I/O** | Windows CP1252 Decode Failure | Byte 0x8f or special Unicode character | Explicit `encoding="utf-8"` on all file handlers | 100% crash prevention across all Windows platforms |
```

6. Telemetry, Tracing & Operational Monitoring

The system implements distributed observability via **Pydantic Logfire** and **LangSmith**:

```
[User Chat Interaction] (UI Span)
    [Calling RAG Backend] (UI -> HTTP)
         [/query Request] (FastAPI Ingress)
              [Layer 1 Guardrail Check]
              [NeMo Guardrails Check] (LLM Span)
              [LangGraph Execution] (State Machine Span)
                   [Planner Decision] (Groq LLM Span)
                   [Knowledge Retrieval] (Qdrant Search Span)
                       [Embed Query] (Gemini / Local Span)
```

```
                    [Semantic Reranking] (FlashRank Cross-Encoder Span)
                [LLM Synthesis] (Portkey / Groq Span)
                     [Gateway Cache Check] (Header Inspection)
```

- **Metrics Tracked:** Request latency, Token counts, Portkey cache hit/miss status, Vector retrieval similarity scores, Cross-encoder semantic scores, and Guardrail classification accuracy.

End of Specification -- Enterprise Agentic RAG System